

# RITIK SAXENA

Dublin, Ireland | +353 89 941 7523 | [ritiksaxena2009@gmail.com](mailto:ritiksaxena2009@gmail.com)  
[linkedin.com/in/ritik-saxena-6aa925176](https://linkedin.com/in/ritik-saxena-6aa925176) | [github.com/ritik90](https://github.com/ritik90)

## PROFESSIONAL SUMMARY

Passionate Data Scientist with 3+ years of hands-on experience in machine learning, natural language processing, and Generative AI, delivering end-to-end solutions spanning data analysis, model development, deployment, and production monitoring. Proven track record of fine-tuning large language models, architecting agentic RAG systems, and building predictive engines that have driven measurable business outcomes - including **€1.5 M+** in monthly incremental revenue. Currently deepening expertise through an MSc in Data Science at Trinity College Dublin, with a focus on advanced AI and data-driven innovation.

## TECHNICAL SKILLS

|                      |                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------|
| Languages            | Python, SQL, C++, PySpark                                                                                             |
| ML / Deep Learning   | TensorFlow, PyTorch, Scikit-learn, SciPy, Hugging Face Transformers, NLTK, SpaCy                                      |
| GenAI / LLMs         | LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine-tuning, Gemini, GPT, BERT, LLama |
| Data & Visualisation | Pandas, NumPy, Plotly, Streamlit                                                                                      |
| MLOps / Deployment   | MLflow, FastAPI, Docker, Evidently, CI/CD Pipelines, REST APIs                                                        |
| Cloud Platforms      | Google Cloud Platform (Vertex AI, Cloud Run, BigQuery), Microsoft Azure (Databricks, App Service, CosmosDB)           |
| Databases            | MongoDB, BigQuery, CosmosDB                                                                                           |
| Developer Tools      | Git, Visual Studio Code, SHAP, A/B Testing Frameworks                                                                 |

## PROFESSIONAL EXPERIENCE

### Data Scientist 2

Oct 2024 - Aug 2025

TVS Motor Company

Bengaluru, India

#### ServiceMate AI - Multi-RAG Agentic System

LangGraph · Gemini LLMs · GPT-4o · Vector Databases (Databricks UC, Azure CosmosDB) · Chainlit · Azure App Service

- Designed and implemented a general-purpose chatbot capable of retrieving brand- and use-case-specific documents using agentic decision-making.
- Developed intelligent agents leveraging the LangGraph orchestration framework and iterative prompting techniques with Gemini and GPT-4o models to ensure accurate, stateful conversation flow.
- Deployed the solution on Azure App Service with a Chainlit-based open-source UI/UX for seamless user interaction.
- Integrated vector search using Databricks UC Vector Store and Azure CosmosDB, enabling fast and contextually relevant document retrieval across multiple brands.

#### Fine-Tuning BERT for Domain-Specific NLP Tasks

BERT · Hugging Face Transformers · PyTorch · Scikit-learn · Python

- Contextual Pretraining & Sentiment Analysis:** Pre-trained BERT with Masked Language Modelling (MLM) on 200 GB+ of domain-specific service data, significantly enhancing contextual understanding for targeted applications.
- Fine-tuned a 3-class sentiment analysis model (Positive, Neutral, Negative), achieving a **96% accuracy rate** on customer interactions.
- Implemented customer clustering based on sentiment, directly improving follow-up efficiency and service team prioritisation by an estimated **25%**.
- Named Entity Recognition (NER):** Developed and fine-tuned a BERT-based NER model on 10,000+ transcribed call recordings, accurately identifying entities such as brands, geographies, and product categories.
- Achieved a robust **F1-score of 94%** for entity extraction, demonstrating high precision and recall.
- Empowered sales and support teams with structured insights from unstructured conversations, leading to **15% faster** tailored follow-ups and improved customer engagement.

## Data Scientist

TVS Motor Company

Aug 2022 - Sept 2024

Bengaluru, India

### Lead Classification Engine

Python · Databricks · MLflow · REST API · Docker · SQL · Scikit-learn · SHAP

- Designed and deployed a real-time and batch machine learning model predicting customer purchase probability, generating **€1.5 M+ in monthly incremental revenue** for 2-wheeler sales across multiple enquiry channels.
- Implemented MLOps workflows in Databricks with MLflow model registry, automated CI/CD pipelines, REST API-based serving, endpoint management, and monitoring.
- Scaled adoption to **200+ dealers**, improving HOT lead follow-up within 2 hours from 30% to **90%**, significantly enhancing lead conversion efficiency.
- Built an integrated monitoring and alerting framework in Databricks to track model, data, system, and business KPIs, ensuring high reliability and incorporating SHAP-based explainability for transparent decision-making.

### Website Hyper-Personalisation

Python · Google Cloud · FastAPI · Docker · Random Forest · XGBoost · LightGBM · AutoML · Vertex AI · SQL

- Built and optimised a multi-label classification model using ensemble methods (Random Forest, XGBoost, LightGBM) to predict client propensity for CTA engagement in real time.
- Performed advanced EDA, feature engineering, and feature selection, leveraging AutoML to identify the most effective classifier.
- Deployed the model on Google Cloud via a FastAPI-based containerised Docker service, validated through load testing to ensure scalable, low-latency real-time scoring.
- Validated effectiveness through rigorous A/B testing, ensuring statistically significant improvements in engagement.
- Implemented a production-grade MLOps pipeline with CI/CD on Google Cloud Run and Vertex AI monitoring for continuous model performance and drift detection.

## PROJECTS

### Search Engine with Automated Evaluation

[GitHub](#)

Java · Apache Lucene · Maven · GitHub Actions · Information Retrieval

- Built a search engine in Java from scratch, implementing document indexing, topic-based retrieval, and TREC-format ranked output using Apache Lucene.
- Set up automated evaluation with GitHub Actions to benchmark retrieval quality using MAP, P@20, and nDCG@20 for continuous testing and performance tracking.

### Spatio-Temporal Audio-Video Analysis for Pain Detection

Python · TensorFlow · NLP · Machine Learning

- Developed a multimodal deep learning system for automatic pain expression detection in infants using spatio-temporal audio-video analysis.
- Extracted audio, visual, and textual features from video and combined their results to predict the emotion of the subject with an accuracy of 64%.

## EDUCATION

### Master of Science in Data Science

Trinity College Dublin

Sept 2025 - Aug 2026

Dublin, Ireland

### Bachelor of Engineering in Computer Science

Thapar Institute of Engineering & Technology

Jul 2018 - Aug 2022

Punjab, India